Tub sonlar, tub ko'paytuvchilarga ajratish va daraja

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Qaysi qatorda 10 dan kichik barcha tub sonlar yozilgan?**

A) 2, 3, 5, 7
B) 3, 5, 7, 9
C) 1, 2, 3, 5, 7
D) 3, 5, 7

2. **100 bilan 110 orasidagi tub sonlar qaysilar?**

A) 103, 107, 109
B) 101, 103, 105, 107, 109
C) 101, 103, 107, 109
D) 101, 103, 107

3. **187 soni tubmi yoki murakkabmi?**

A) Tub: u 2, 3, 5, 7 ga bo'linmaydi
B) Murakkab: 187 = 7 · 27
C) Tub: oxirgi raqami 7
D) Murakkab: 187 = 11 · 17

4. **Qaysi son tub son?**

A) 169
B) 221
C) 173
D) 133

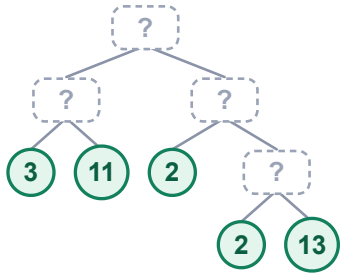
5. **792 ning tub ko'paytuvchilarga yoyilmasini daraja shaklida yozing.**

A) $2^2 \cdot 3^3 \cdot 11$
B) $2^3 \cdot 3 \cdot 11$
C) $8 \cdot 9 \cdot 11$
D) $2^3 \cdot 3^2 \cdot 11$

6. $2^3 \cdot 11$ **qaysi songa teng?**

A) 88
B) 55
C) 66
D) 1 331

7. **Ko'paytuvchilar daraxtidagi bo'sh joylarni to'ldiring. Tepadagi son nechaga teng?**



A) 858
B) 85
C) 572
D) 1 716

8. $60 \cdot w$ **ko'paytma 45 ga karrali bo'lishi uchun** w **natural sonning eng kichik qiymati qancha?**

A) 9
B) 3
C) 45
D) 15

9. $18 \cdot n$ **ko'paytma 100 ga karrali bo'lishi uchun** n **natural sonning eng kichik qiymati qancha?**

A) 25
B) 100
C) 10
D) 50

10. **45 ta konfetni bir nechta bolaga teng bo'lib berish kerak: bolalar 1 tadan ko'p, har bolaga ham 1 tadan ko'p konfet tegsin. Necha xil usulda bo'lish mumkin?**

A) 4 **B)** 3 **C)** 2 **D)** 6

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11. **52 ning nechta bo'luvchisi bor?**

A) 3 **B)** 5 **C)** 4 **D)** 6

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12. **Son $3^2 \cdot 5 \cdot 7$ shaklida tub ko'paytuvchilarga ajratilgan. Qaysi son uning bo'luvchisi emas?**

A) 63 **B)** 35 **C)** 45 **D)** 25

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13. **Uchta har xil tub sonning ko'paytmasi 105 ga teng. Shu uch sonning yig'indisi nechaga teng?**

A) 21 **B)** 12 **C)** 15 **D)** 35

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14. **Qaysi son 4 ga ham, 10 ga ham bo'linadi, lekin 40 ga bo'linmaydi?**

A) 50 **B)** 30 **C)** 80 **D)** 20

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15. **4^3 va 3^4 ni solishtiring.**

A) teng, ikkalasi 12 ga teng **B)** teng, ikkalasi 7 ga teng

C) 3^4 katta **D)** 4^3 katta

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16. **2 ning qaysi darajasi 128 ga teng?**

A) 7 **B)** 64 **C)** 6 **D)** 8

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17. **Rasmda 1 650 ni qisqa bo'lish boshlangan. Oxirigacha davom ettirsak, yoyilma qanday bo'ladi?**

1650	2
825	3
275	

A) $2 \cdot 3 \cdot 275$ **B)** $2 \cdot 3 \cdot 5^2 \cdot 11$

C) $2 \cdot 3 \cdot 5 \cdot 11$ **D)** $2 \cdot 3 \cdot 5^2 \cdot 13$

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18. **$2 \cdot 3^2 \cdot p = 306$, bunda p — tub son. p ni toping.**

A) 51 **B)** 153 **C)** 17 **D)** 34

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19. **Madina 30 bilan 50 orasida son o'yladi. Uning tub ko'paytuvchilari faqat 2 va 3 (ikkalasi ham bor), 2 lar soni 3 lar sonidan ko'p. Madina qaysi sonni o'yladi?**

A) 32 **B)** 48 **C)** 42 **D)** 36

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20. **Aziz 132 ni tub ko'paytuvchilarga ajratib, 132 = 4 · 3 · 11 deb yozdi. To'g'ri yoyilma qaysi?**

A) Aziz to'g'ri yozgan **B)** $2^2 \cdot 3 \cdot 11$

C) $2^2 \cdot 33$ **D)** $2 \cdot 3 \cdot 11$

Javoblar

Tub sonlar, tub ko'paytuvchilarga ajratish va daraja · 2-test (qiyinroq)

№	Javob	Qiymati
1	**A**	2, 3, 5, 7
2	**C**	101, 103, 107, 109
3	**D**	Murakkab: 187 = 11 · 17
4	**C**	173
5	**D**	$2^3 \cdot 3^2 \cdot 11$
6	**A**	88
7	**D**	1 716
8	**B**	3
9	**D**	50
10	**A**	4

№	Javob	Qiymati
11	**D**	6
12	**D**	25
13	**C**	15
14	**D**	20
15	**C**	3^4 katta
16	**A**	7
17	**B**	$2 \cdot 3 \cdot 5^2 \cdot 11$
18	**C**	17
19	**B**	48
20	**B**	$2^2 \cdot 3 \cdot 11$

Qisqa yechimlar

1. 1 tub emas (bo'luvchisi bitta), 9 = 3 · 3 — murakkab.
2 juft bo'lsa ham tub.
Javob: **2, 3, 5, 7**.

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2. Juft sonlar va 105 (5 bilan tugaydi) tushib qoladi. Qolganlari: 101, 103, 107, 109.
Ularni 3 va 7 ga tekshiramiz: hech biri bo'linmaydi.
Javob: **101, 103, 107, 109**.

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3. 187 son 2, 3, 5, 7 ga bo'linmaydi — lekin bu hali yetarli emas.
Keyingi tub son 11: 187 : 11 = 17.
Javob: **murakkab, 187 = 11 · 17**. (7 · 27 = 189.)

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4. 133 = 7 · 19, 169 = 13 · 13, 221 = 13 · 17.
173 esa 2, 3, 5, 7, 11, 13, … — o'zidan kichik birorta tub songa qoldiqsiz bo'linmaydi.
Javob: **173**.

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5. 792 → 396 → 198 → 99 → 33 → 11 → 1 (2, 2, 2, 3, 3, 11).
792 = $2^3 \cdot 3^2 \cdot 11$.
8 va 9 tub emas; $2^3 \cdot 3 \cdot 11 = 264$.

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6. $2^3 = 2 \cdot 2 \cdot 2 = 8$ (6 emas!).
8 · 11 = **88**.
66 — 2^3 ni 2 · 3 deb hisoblansa chiqadi.

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7. Tepadagi son — daraxtdagi barcha tub sonlar ko'paytmasi: 3 · 11 · 2 · 2 · 13.
3 · 11 = 33, 2 · 2 · 13 = 52, 33 · 52 = **1 716**.
858 — bitta 2 tushib qolgan.

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8. $45 = 3^2 \cdot 5$, $60 = 2^2 \cdot 3 \cdot 5$.
60 da 5 bor, 3 esa bitta — yana bitta 3 yetishmaydi.
w = **3**. Tekshirish: 60 · 3 = 180, 180 : 45 = 4.

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9. $100 = 2^2 \cdot 5^2$, $18 = 2 \cdot 3^2$.
18 da bitta 2 bor; yana bitta 2 va ikkita 5 kerak: $n = 2 \cdot 5^2$ = **50**.
25 yetmaydi: 18 · 25 = 450, 100 ga bo'linmaydi.

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10. 45 ning bo'luvchilari: 1, 3, 5, 9, 15, 45.
1 va 45 shartga to'g'ri kelmaydi; qoladi 3, 5, 9, 15 ta bola.
Javob: **4** usul (masalan, 3 bolaga 15 tadan yoki 15 bolaga 3 tadan).

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11. 52 = 1 · 52 = 2 · 26 = 4 · 13.
Bo'luvchilar: 1, 2, 4, 13, 26, 52 — **6** ta.
1 ni va 52 ning o'zini ham sanash kerak.

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12. 45 = $3^2 \cdot 5$, 35 = 5 · 7, 63 = $3^2 \cdot 7$ — hammasi yoyilmadagi ko'paytuvchilardan tuzilgan.
25 = 5^2, yoyilmada esa 5 bitta.
Javob: **25**.

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13. 105 → 35 → 7 → 1: 105 = 3 · 5 · 7.
Yig'indi: 3 + 5 + 7 = **15**.

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14. 20 : 4 = 5, 20 : 10 = 2, lekin 20 son 40 ga bo'linmaydi.
Javob: **20**. Demak, «4 ga va 10 ga bo'linsa, 40 ga ham bo'linadi» degan gap yolg'on.
80 son 40 ga bo'linadi; 30 va 50 esa 4 ga bo'linmaydi.

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15. $4^3 = 4 \cdot 4 \cdot 4 = 64$, $3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81$.
81 > 64, demak 3^4 **katta**.
4 · 3 = 12 deb hisoblash — xato.

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16. 2, 4, 8, 16, 32, 64, 128 — har safar 2 ga ko'paytiramiz.
128 da 2 soni 7 marta ko'paytirilgan: $2^7 = 128$.
Javob: **7**.

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17. 275 → 55 → 11 → 1: 275 ni ikki marta 5 ga, keyin 11 ga bo'lamiz.
1 650 = 2 · 3 · 5 · 5 · 11 = $2 \cdot 3 \cdot 5^2 \cdot 11$.
275 — murakkab, u yoyilmada qolmaydi.

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18. $2 \cdot 3^2 = 18$ (2 · 6 = 12 emas).
p = 306 : 18 = **17** — tub son.
34 = 306 : 9 (2 unutilgan), 51 = 306 : 6 (bitta 3 olingan) — ular tub ham emas.

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19. Faqat 2 va 3 dan tuzilgan sonlar: 32 = 2^5 (3 yo'q), 36 = $2^2 \cdot 3^2$ (teng), 48 = $2^4 \cdot 3$.
42 = 2 · 3 · 7 — 7 bor.
Javob: **48**.

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20. 4 — murakkab son, 4 = 2 · 2.
132 = 2 · 2 · 3 · 11 = $2^2 \cdot 3 \cdot 11$.
2 · 3 · 11 = 66 — bitta 2 tushib qolgan; 33 ham tub emas.
